1.3.5 Sizing or Starching

Size is one of the numerous substances that is applied to a yarn or fabric to act as a protective filler or glaze. In sizing or starching, the fabric is immersed in a mixture containing waxes, oils, glycerines and softeners to control the fabric body by adding stiffness and weight. If the sizing is resin based and heat set, it is permanent in nature but if the size is water soluble, it is removed during washing. Gelatin is used on rayon's because it is a clear substance that enhances the natural lustre of fibres.



Figure 1.5 Sizing Machine

In the weaving process the fabric warp has to undergo several types of actions like cyclic strain, flexing, abrasion at various loom parts and inter yarn friction. To reduce the breakage of yarns in the weaving process the warp yarns are sized before weaving which increases its strength, abrasion resistance and decreases the yarn hairiness. Different types of water soluble Polymers Vinyl Alcohol (PVA) called textile sizing agents or chemicals such as modified starch, Polyvinyl Carboxy Methyl Cellulose (CMC) and acrylates are used as sizing agents. In order to reduce the abrasiveness of the warp yarns wax is also added to the sizing liquor. The type of yarn material, the thickness of the yarn, type of weaving machinery determines the sizing recipe. The sizing can be done by hand or sizing machine. After the weaving process gets over the fabric is desized to remove the sizing liquor.

1.3.6 Stiffening

Stiffening agents are applied to the cloth to increase the weight of the fabric, to improve its thickness and lustre. Depending on the end use of the fabric, some fabrics are needed to be made stiffer and crispier. But, the effect of these stiffening agents is temporary and once the fabric is washed, most of the finishes are removed. Stiffening agents such as starches which are used for finishing of cotton cloth are derived from potato, wheat or corn. Dextrin's are used for dyed and printed fabrics as they do not have any undue effect on the dye or print of the cloth. Natural gums are mainly used in printing as well as finishing process whereas modified cellulose like resins are used as stiffening agents.



Figure 1.6 Double Stiffened Buckram Fabric

Acid stiffening is mostly used for fine yarn cotton fabrics as it gives stiffness as well as transparency to the fabric. It involves rapid immersion in sulphuric acid, followed by immediate neutralization

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